

Figure 1: Causal graphs of four datasets. (a) shows the causal graph of Age dataset of CelebA. (b) shows the causal graph of Smile dataset of CelebA. (c) shows the causal graph of Pend synthetic dataset. (d) shows the causal graph of Flow synthetic dataset.

Theory Proof of Equation

In this section, we show the details of conversion from Equation 1 to Equation 2 in the main text. The Equation 1 can be converted as follows:

$$\begin{aligned}
& \mathcal{D}_{KL}(q(z|x, u), p(z|x, u)) \\
&= \int q(z|x, u) (\log q(z|x, u) - \log p(z|x, u)) dz \\
&= \mathbb{E}_{z \sim q(z|x, u)} [\log q(z|x, u) - \log p(z|x, u)] \\
&= \mathbb{E}_{z \sim q(z|x, u)} [\log q(z|x, u) - \log p(x|z, u) - \log p(z|u)] \\
&\quad + \log p(x|u)
\end{aligned} \tag{1}$$

In this way, we can obtain:

$$\begin{aligned}
& -\log p(x|u) + \mathcal{D}_{KL}(q(z|x, u), p(z|x, u)) \\
&= \mathbb{E}_{z \sim q(z|x, u)} [\log q(z|x, u) - \log p(z|u) - \log p(x|z, u)] \\
&= -\mathbb{E}_{z \sim q(z|x, u)} [\log p(x|z, u)] + \mathcal{D}_{KL}(q(z|x, u), p(z|u))
\end{aligned} \tag{2}$$

where $q(z|x, u)$ represents extracting latent representations z from the image x through the encoder and injecting causal effects of annotations u into z through SCM (Pearl 2009), $p(x|z, u)$ represents reconstructing the image x with latent representations possessing causal effects in u as input through the decoder and $p(z|u)$ represents z are conditioning on u .

Considering the encoder, SCM and decoder, Equation 2 can be rewritten as:

$$\begin{aligned}
\mathcal{L} &= -\log p(x|u) + \mathcal{D}_{KL}(q(z|x, u), p(z|x, u)) \\
&= -\mathbb{E}_{z \sim q_{E, F}(z|x, u)} [\log p_D(x|z, u)] \\
&\quad + \mathcal{D}_{KL}(q_{E, F}(z|x, u), p_\lambda(z|u)) \\
&= \mathcal{L}_{rec} + \mathcal{L}_{kl}
\end{aligned} \tag{3}$$

where E is the encoder, F is causal functions in nonlinear SCM, D is the decoder and λ are functions that z conditioning on u . To express easily latter, we define the first term in $\mathcal{L}$ as $\mathcal{L}_{rec}$ and the second term in $\mathcal{L}$ as $\mathcal{L}_{kl}$.

Identifiability Analysis Supplement

In this section, we give the detailed proof of ψ in the main text as a mapping proving CDM equivalence. The general

logic of proof follows (Yang et al. 2021). For this, we first review the mapping ψ as:

$$\psi : \begin{cases} T(E(x)) = L_1 \tilde{T}(\tilde{E}(x)) + l_1 : \epsilon \rightarrow \tilde{\epsilon} \\ T(D^{-1}(x)) = L_2 \tilde{T}(\tilde{D}^{-1}(x)) + l_2 : \mathcal{Z} \rightarrow \tilde{\mathcal{Z}} \end{cases} \tag{4}$$

where E is the encoder, D is the decoder, T is the sufficient statistics for latent representations z that $T \neq 0$, ϵ are exogenous variables, $\sim$ denotes the learned parameters, L_1 is an invertible matrix, L_2 is an invertible diagonal matrix with diagonal elements associated to u , l_1 and l_2 are vectors, the decoder function D is differentiable and the Jacobian matrix of D is of full rank.

We begin the proof with $p_\theta(x|u) = p_{\tilde{\theta}}(x|u)$. The equation can be converted as:

$$\begin{aligned}
& \int_z p_D(x|z) p_\lambda(z|u) dz = \int_z p_{\tilde{D}}(x|z) p_{\tilde{\lambda}}(z|u) dz \Rightarrow \\
& \int_{x'} p_D(x|D^{-1}(x')) p_\lambda(D^{-1}(x')|u) |det(J_{D^{-1}}(x'))| dx' \\
&= \int_{x'} p_{\tilde{D}}(x|\tilde{D}^{-1}(x')) p_{\tilde{\lambda}}(\tilde{D}^{-1}(x')|u) |det(J_{\tilde{D}^{-1}}(x'))| dx'
\end{aligned} \tag{5}$$

For decoder function D , $p_D(x|D^{-1}(x'))$ is equivalent to the Gaussian distribution $p_\eta(x - x')$ and this term can be eliminated (Yang et al. 2021). Thus the Equation 5 is transformed as:

$$\begin{aligned}
& p_\lambda(D^{-1}(x')|u) |det(J_{D^{-1}}(x'))| \\
&= p_{\tilde{\lambda}}(\tilde{D}^{-1}(x')|u) |det(J_{\tilde{D}^{-1}}(x'))|
\end{aligned} \tag{6}$$

Considering the sufficient statistics T , where $T(\cdot) = (\mu(\cdot), \sigma(\cdot))$ and conditioning functions λ , where $\lambda = \text{diagonal}(\lambda_1(u_1), \dots, \lambda_n(u_n))$, like (Yang et al. 2021), we obtain:

$$\begin{aligned}
& \log |det(J_{D^{-1}}(x))| - \log \sigma_z(D^{-1}(x)) + \log z_u \\
&+ \sum_{i=1}^2 T_i(D^{-1}(x)) \lambda_i(u) \\
&= \log |det(J_{\tilde{D}^{-1}}(x))| - \log \tilde{\sigma}_z(\tilde{D}^{-1}(x)) + \log \tilde{z}_u \\
&+ \sum_{i=1}^2 \tilde{T}_i(\tilde{D}^{-1}(x)) \tilde{\lambda}_i(u)
\end{aligned} \tag{7}$$

Since $\tilde{A}$ as DAG and $(I - \tilde{A}^T)^{-1}$ as full rank, Equation 7 is rewritten as:

$$\sum_{i=1}^2 T_i(D^{-1}(x)) \lambda_i(u) = \sum_{i=1}^2 \tilde{T}_i(\tilde{D}^{-1}(x)) \tilde{\lambda}_i(u) + l_u \tag{8}$$

Define $Q = (I - A^T)^{-1}$, Equation 8 can be rewritten as:

$$\sum_{i=1}^2 T_i(Q \cdot E(x)) \lambda_i(u) = \sum_{i=1}^2 \tilde{T}_i(\tilde{Q} \cdot \tilde{E}(x)) \tilde{\lambda}_i(u) + l'_u \tag{9}$$

Assuming the supervision annotations are distinct, there is an invertible matrix $L = \text{diagonal}(\lambda_1(u), \lambda_2(u))$ (Yang et al. 2021), that:

$$L \cdot T(D^{-1}(x)) = \tilde{L} \cdot \tilde{T}(\tilde{D}^{-1}(x)) + l \tag{10}$$

encoder	decoder
4×4 conv-32, stride 2, padding 1, LeakyReLU(0.2) 4×4 conv-32, stride 2, padding 1, LeakyReLU(0.2) 4×4 conv-64, stride 2, padding 1, LeakyReLU(0.2) 4×4 conv-64, stride 2, padding 1, LeakyReLU(0.2) 4×4 conv-64, stride 2, padding 1, LeakyReLU(0.2) 4×4 conv-256, stride 1, LeakyReLU(0.2) 1×1 conv-544, stride 1 -	1×1 conv-256, LeakyReLU(0.2) 4×4 convtranspose-128, LeakyReLU(0.2) 4×4 convtranspose-128, stride 2, padding 1, LeakyReLU(0.2) 4×4 convtranspose-64, stride 2, padding 1, LeakyReLU(0.2) 4×4 convtranspose-64, stride 2, padding 1, LeakyReLU(0.2) 4×4 convtranspose-64, stride 2, padding 1, LeakyReLU(0.2) 4×4 convtranspose-32, stride 2, padding 1, LeakyReLU(0.2) 1×1 convtranspose-3, stride 1

Table 1: Encoder-Decoder for CelebA dataset of CFI-VAE.

encoder	decoder
$4 * 96 * 96 \times 900$ fc., ELU 900×300 fc., ELU $300 \times 2 * concepts * 4$ fc. -	$concepts \times (4 \times 300$ fc., ELU) $concepts \times (300 \times 300$ fc., ELU) $concepts \times (300 \times 1024$ fc., ELU) $concepts \times (1024 \times 4 * 96 * 96$ fc.)

Table 2: Encoder-Decoder for synthetic datasets of CFI-VAE.

And this can be rewritten as:

$$T(D^{-1}(x)) = L_2 \cdot \tilde{T}(\tilde{D}^{-1}(x)) + l_2 : z \rightarrow \tilde{z} \quad (11)$$

where $L_2 = \lambda(u)^{-1} \cdot \tilde{\lambda}(u)$

Similar Equation 11, we can obtain:

$$T(E(x)) = L_1 \cdot \tilde{T}(\tilde{E}(x)) + l_1 : \epsilon \rightarrow \tilde{\epsilon} \quad (12)$$

In this way, we could prove that ψ is the mapping makes CDM equivalence $\mathcal{M}_\theta \sim \tilde{\mathcal{M}}_{\tilde{\theta}}$.

More Experimental Details

Causal Graph Visualization

In this section, we visualize the causal graphs for real-world and synthetic datasets in the main text. Like (Yang et al. 2021), the causal graphs for Age, Smile, Pend and Flow datasets are shown in Figure 1.

Data Preprocessing

The original resolution of CelebA image is at 218×178 , which contains face, body and background. Because we only focus on face image containing causal concepts of interest, we need to preprocess original CelebA image. Inspired by CausalGAN (Kocaoglu et al. 2017), we preprocess original CelebA image through two steps. We first transfer CelebA images to 108×108 resolution by centrally cropping with random horizontal flip, then we resize them at 128×128 resolution.

More Training Details

We use NVIDIA Quadro RTX 6000 GPU with 24GB memory as our experiment device. The experiments are conducted on Linux OS. For the metrics, MIC and TIC (Kinney and Atwal 2014) in the main text are estimated by computing score function in class MINE of minepy in Pytorch.

Architecture Details

Counterfactual intervention module. Core of this module is a classifier, which is composed by linear neural network. The classifier is made up from three linear layers, whose output dimensions are 64, 128 and category number respectively. The classifier is used to estimate causal effect (prediction logits) from one variable to another variable.

The encoder-decoder architecture. **1)** The architecture of the encoder-decoder on CelebA dataset (Liu et al. 2015) (Age and Smile datasets) is shown in Table 1. The encoder is made up from seven convolutional layers, each one with LeakyReLU activation function. The first five convolutions are performed with a stride 2 and a padding 1. The encoder is used to transfer an input image at 128×128 resolution to exogenous variables expressed by a $concepts \times 68$ vector. The first $concepts \times 64$ vector represents independent representations corresponding to the physical concepts of interest and the rest of vector represents the other information in the image. The decoder is made up from one convolutional layer and seven anti-convolutional layers, each one with LeakyReLU activation function. The middle five anti-convolutions are performed with a stride 2 and a padding 1. The decoder is built to transfer latent representations, which is a $concepts \times 68$ vector, to reconstructed images at 128×128 resolution. **2)** The architecture of the encoder-decoder on synthetic datasets (Pend and Flow datasets) is shown in Table 2. The encoder is made up from three linear layers with ELU activation function. The encoder is used to transfer an input image at 96×96 resolution to exogenous variables expressed by a $concepts \times 16$ vector. Because the image in synthetic datasets only possesses elements corresponding to the physical concepts of interest, the vector only represents physical representations. The decoder is made up from four linear layers with ELU activation function. The decoder is built to transfer latent representations, which is a

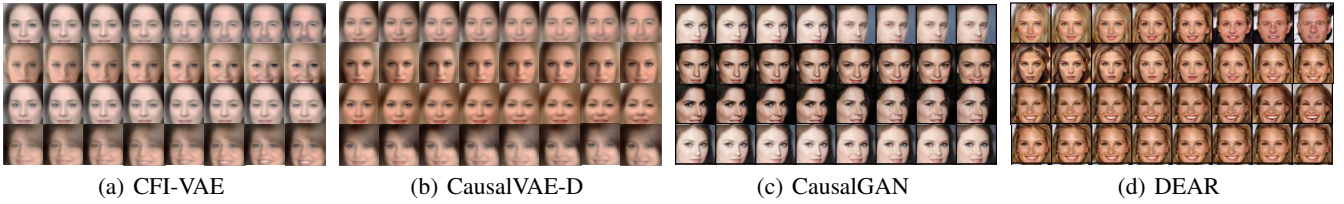


Figure 2: Intervention results of CFI-VAE, CausalVAE-D, CausalGAN and DEAR on Smile dataset, where the intervention factors are *gender*, *smile*, *eye* and *mouth* respectively from up to bottom.

Method	MIC $\uparrow$				TIC $\uparrow$				ISR(%) $\uparrow$				CERD(%) $\uparrow$				SCR(%) $\downarrow$			
	Age	Smile	Pend	Flow	Age	Smile	Pend	Flow	Age	Smile	Pend	Flow	Age	Smile	Pend	Flow	Age	Smile	Pend	Flow
CausalVAE	0.277	0.392	0.588	0.430	2.149	3.008	5.693	3.632	25	25	100	75	0	25	100	67	50	38	75	56
CausalVAE-D	0.322	0.414	0.643	0.433	2.481	3.263	5.752	3.602	100	50	100	50	75	75	0	67	63	38	25	33
CausalGAN	0.186	0.103	-	-	0.522	0.591	-	-	75	100	-	-	25	25	-	13	38	-	-	-
DEAR	0.218	0.280	0.228	0.238	1.204	1.585	1.253	1.620	25	50	100	50	25	0	100	33	50	25	100	44
CFI-VAE	0.336	0.466	0.699	0.787	2.646	3.894	6.306	7.511	100	100	100	75	75	100	100	67	13	25	25	22

Table 3: Comparison with the state-of-the-arts on Age, Smile, Pend and Flow datasets under proportion of labeled samples as 1%. Best performances are bold.

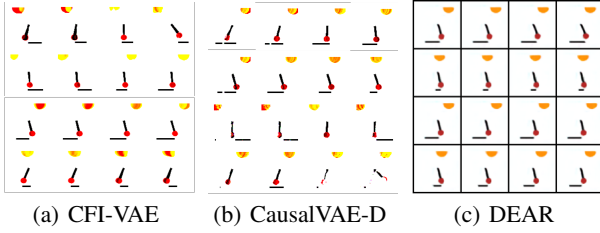


Figure 3: Intervention results of CFI-VAE, CausalVAE-D and DEAR on Pend dataset, where the intervention factors are *pendulum angle*, *light position*, *shadow length* and *shadow position* respectively from up to bottom.

$concepts \times 16$ vector, to reconstructed images at 96×96 resolution. We all follow and improve the neural network design strategy of CausalVAE (Yang et al. 2021).

Additional Experimental Results

Additional Qualitative Results

In this section, we exhibit more qualitative results on Smile dataset of CelebA (Liu et al. 2015), and Pend and Flow synthetic datasets.

For Smile dataset, the results are shown in Figure 2. **1)** For our CFI-VAE, we can observe that when *gender* is intervened, *eye* changes, that when *smile* is intervened, *eye* and *mouth* change, and that when *mouth* is intervened, *eye* changes. **2)** For CausalVAE-D, we can observe that intervention on *smile* cannot change *mouth*, that intervention on *eye* changes *smile* and that intervention on *mouth* changes *smile*. **3)** For CausalGAN, intervention on *smile* cannot change *eye* and intervention on *mouth* changes *smile*. **4)** For DEAR, intervention on *gender* or *eye* changes *smile* and *mouth*, and *mouth* is not intervened successfully.

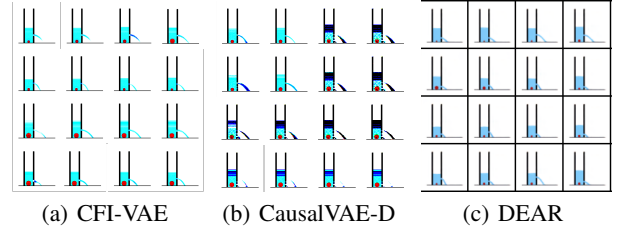


Figure 4: Intervention results of CFI-VAE, CausalVAE-D and DEAR on Flow dataset, where the intervention factors are *ball size*, *water height*, *water flow* and *hole location* respectively from up to bottom.

For Pend dataset, the results are shown in Figure 3, except for CausalGAN because of the model setup constraint on binary classification datasets. **1)** For our CFI-VAE, we can observe that when *pendulum angle* is intervened, *shadow length* and *shadow position* change and that when *light position* is intervened, *shadow length* and *shadow position* change. **2)** For CausalVAE-D, intervention on *shadow length* changes *pendulum angle*, and intervention on *shadow position* changes *pendulum angle* and *shadow length*. **3)** For DEAR, intervention on *shadow position* changes *light position* and *shadow length*.

For Flow dataset, the results are shown in Figure 4, except for CausalGAN because of the model setup constraint on binary classification datasets. **1)** For our CFI-VAE, we can observe that when *ball size* is intervened, *water height* changes, that when *water height* is intervened, *water flow* changes and that when *hole location* is intervened, *water flow* changes. **2)** For CausalVAE-D, intervention on *water height* changes *ball size*, and *hole location* cannot be intervened successfully. **3)** For DEAR, intervention on *ball size* changes *hole location*, intervention on *water height* changes

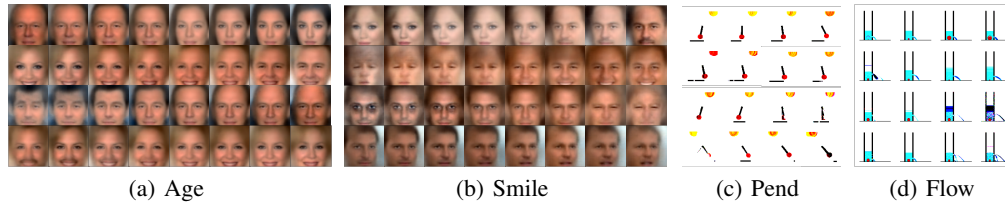


Figure 5: Intervention results of CausalVAE on Age, Smile, Pend and FLOW datasets.

Method	MIC $\uparrow$				TIC $\uparrow$				ISR(%) $\uparrow$				CERD(%) $\uparrow$				SCR(%) $\downarrow$			
	Age	Smile	Pend	Flow	Age	Smile	Pend	Flow	Age	Smile	Pend	Flow	Age	Smile	Pend	Flow	Age	Smile	Pend	Flow
CausalVAE	0.117	0.125	0.384	0.397	0.418	0.483	2.776	2.440	50	50	100	50	50	0	100	33	25	25	50	56
CausalVAE-D	0.110	0.125	0.394	0.374	0.428	0.506	2.823	2.319	50	25	25	50	50	0	0	33	63	38	13	56
CausalGAN	0.061	0.058	-	-	0.345	0.371	-	-	50	75	-	-	0	25	-	-	13	38	-	-
DEAR	0.119	0.148	0.186	0.164	0.434	0.657	0.861	0.736	25	75	50	25	25	50	25	0	38	25	63	67
CFI-VAE	0.125	0.188	0.486	0.515	0.499	1.320	3.517	3.557	50	75	50	50	50	100	100	33	13	25	25	11

Table 4: Comparison with the state-of-the-arts on Age, Smile, Pend and Flow datasets under proportion of labeled samples as 0.1%. Best performances are bold.

ball size and hole location, and intervention on water flow or hole location changes water height.

For the integrity of qualitative experiments, the results of CausalVAE (Yang et al. 2021) are shown in Figure 5. 1) For Age dataset, intervention on age changes gender, intervention on gender cannot change beard, intervention on bald changes age, and intervention on beard changes bald. 2) For Smile dataset, intervention on smile changes gender, and intervention on eye changes mouth. 3) For Pend dataset, intervention on shadow length changes pendulum angle, and intervention on shadow position changes shadow length. 4) For Flow dataset, intervention on ball size cannot change water height, and intervention on water height changes ball size and hole location, and intervention on water flow changes water height, and intervention on hole location changes ball size and water height.

All intervention results of CFI-VAE are strictly conform to the causal graphs set in advance, while all spurious results of baselines mentioned above violate the causal graphs set in advance.

Additional Ablation Analysis for Proportion of Supervision Annotations

In this section, we show results for supervision proportion as 1% and 0.1% as a supplement to ablation study for proportion of supervision annotations in the main text. The results of 1% supervision and 0.1% supervision are summarized in Table 3 and Table 4 respectively. MIC and TIC are 0.572 and 5.089, 0.329 and 2.223 on average for supervision proportion of 1% and 0.1% respectively for CFI-VAE. For CFI-VAE and baseline methods, the decline of MIC and TIC leads that model cannot intervene representations successfully (ISR), which illustrates that disentanglement performance is weakened greatly as supervision proportion reduces. Besides, the significant decline of CERD and the sharp increasing of SCR illustrate that models cannot endow

latent representations with causality of causal graphs set in advance well. For almost all metrics on different datasets, CFI-VAE outperforms baseline methods in causally disentangling for supervision proportion of 1% and 0.1%. In summary, these results illustrate that full proportion of supervision is still needed for causally disentangling task strictly conforming to causal graph set in advance.

References

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